Pictures Become Numbers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Pictures Become Numbers before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Input, Process, Output

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Pictures Become Numbers trace and record the first step you can explain without help.

Use W2: Preserve the missing field for Images as numerical **grid**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Pictures Become Numbers explanation against the supplied facts before

> discussing it.

Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Pictures Become Numbers target to a trace and a justified conclusion.

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Pictures Become Numbers case and show the deciding evidence.

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 5 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	

Test record	My evidence
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Pictures Become Numbers adds to Helpful Classroom Sorter and keep the evidence.

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read and reconstruct

> Read and reconstruct. Apply the supplied Pictures Become Numbers facts and keep your reasoning.

Using 1 for black and 0 for white, draw width 3, height 2 with rows 1,0,1 and 0,1,0. Read it back into numbers and count black **pixel**s.

> **My reasoning and evidence**

W2 Preserve the missing field

> Preserve the missing field. Apply the supplied Pictures Become Numbers facts and keep your reasoning.

A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

> **My reasoning and evidence**

W3 Test a count feature

> Test a count feature. Apply the supplied Pictures Become Numbers facts and keep your reasoning.

A classifier receives only the number of black **pixel**s. Can it always distinguish the chapter's cross from every other three by three pattern? Construct a different pattern with the same count and explain.

> **My reasoning and evidence**

W4 Design a new counterexample

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Design a new counterexample. Apply the supplied Pictures Become Numbers facts and keep your reasoning.

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Locate an input problem

> Locate an input problem. Apply the supplied Pictures Become Numbers facts and keep your reasoning.

A program receives the correct **grid** but displays black wherever the code says 0 and white wherever it says 1. A student proposes collecting more classifier examples.

Identify the earlier mismatch and the first check to perform.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Pictures Become Numbers

> Two fictional 2-by-3 pixel badges contain the same number of dark squares. Explain the labelled step before checking the changed condition.

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

Dark=1, light=0; rows are read top to bottom, left to right.

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/5

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Pixel convention

Each image has two rows and three columns; 1 means dark.

2. Badge A

Top 101, bottom 010: three dark pixels.

3. Badge B

Top 111, bottom 000: also three dark pixels.

4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Find a position where the original badges differ, then read the changed A back as a picture.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The totals agree, but middle-top and middle-bottom positions differ.

1. Compare arrangements

2. Badge B

3. **Pixel** convention

4. Badge A

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare arrangements The totals agree, but middle-top and middle-bottom positions differ. A **grid** preserves where each brightness belongs.

Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change A's bottom-left **pixel** from 0 to 1. Which conclusion is

supported by the supplied facts?

1. A now has four dark **pixel**s; its row **encoding** is 101 / 110.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A now has four dark **pixel**s; its row **encoding** is 101 / 110. Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Repair a missing building block

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Pictures Become Numbers picture from its labels, then record a relevant limit.

01. Pictures Become Numbers: the case

AI ACADEMY / AI-1/5 · 01 · LOOK AT THE FACTS

Pictures Become Numbers: the case

The situation

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

Supplied evidence

Dark=1, light=0; rows are read top to bottom, left to right.
Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

By the end, I can encode pixel positions and show why a count cannot reconstruct a picture.

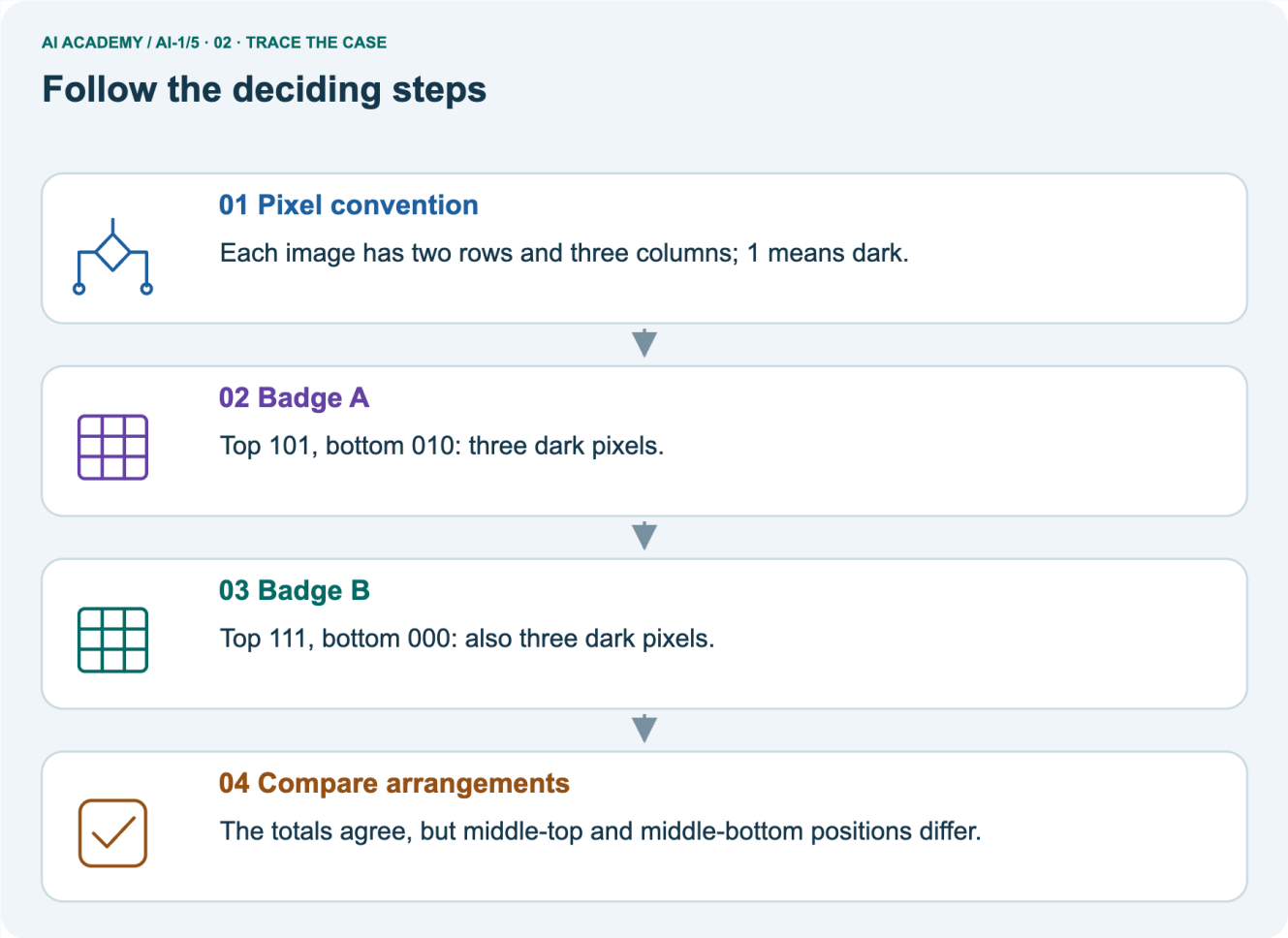
Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares. Supplied facts: Dark=1, light=0; rows are read top to bottom, left to right. Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



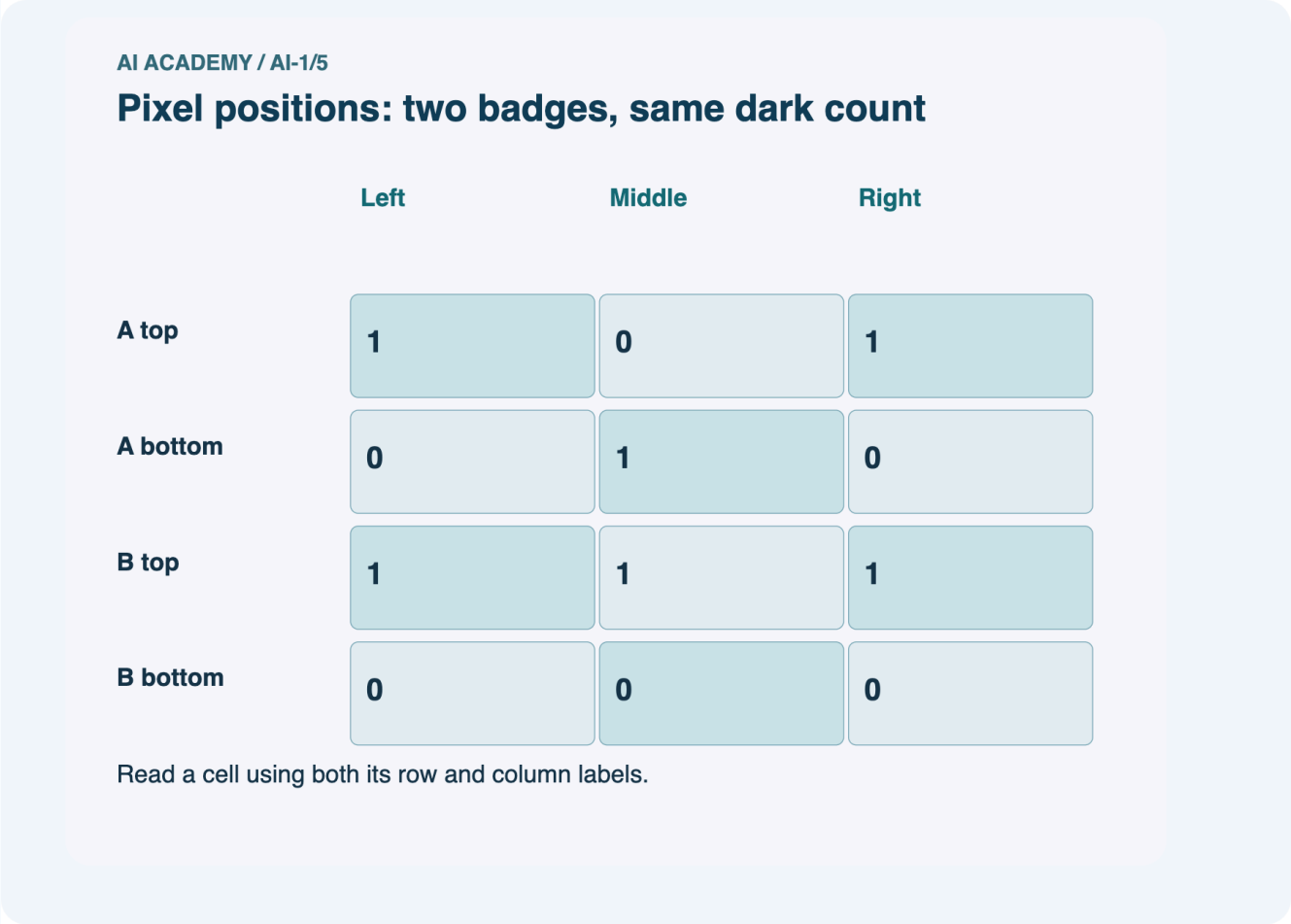
Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel convention: Each image has two rows and three columns; 1 means dark. → Badge A: Top 101, bottom 010: three dark **pixel**s. → Badge B: Top 111, bottom 000: also three dark **pixel**s. → Compare arrangements: The totals agree, but middle-top and middle-bottom positions differ.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Pixel positions: two badges, same dark count



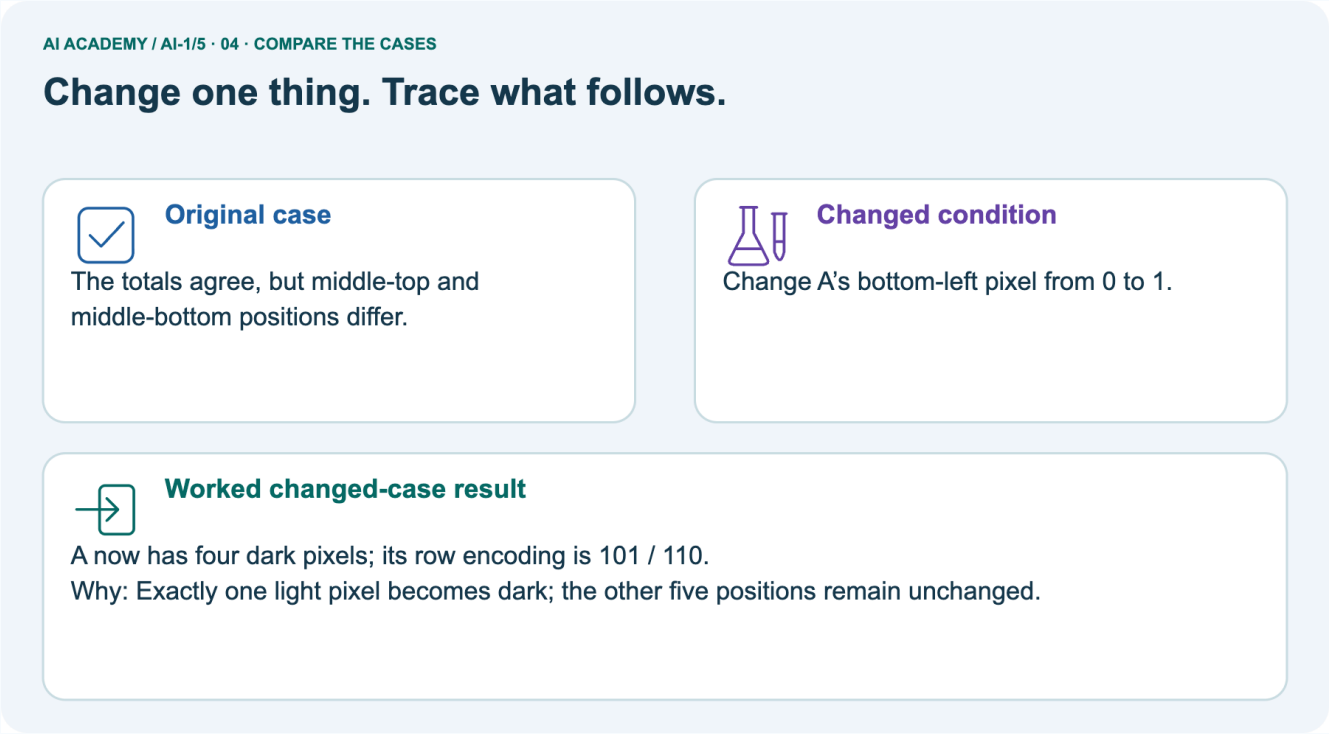
Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel positions: two badges, same dark count. A top: Left = 1, Middle = 0, Right = 1. A bottom: Left = 0, Middle = 1, Right = 0. B top: Left = 1, Middle = 1, Right = 1. B bottom: Left = 0, Middle = 0, Right = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

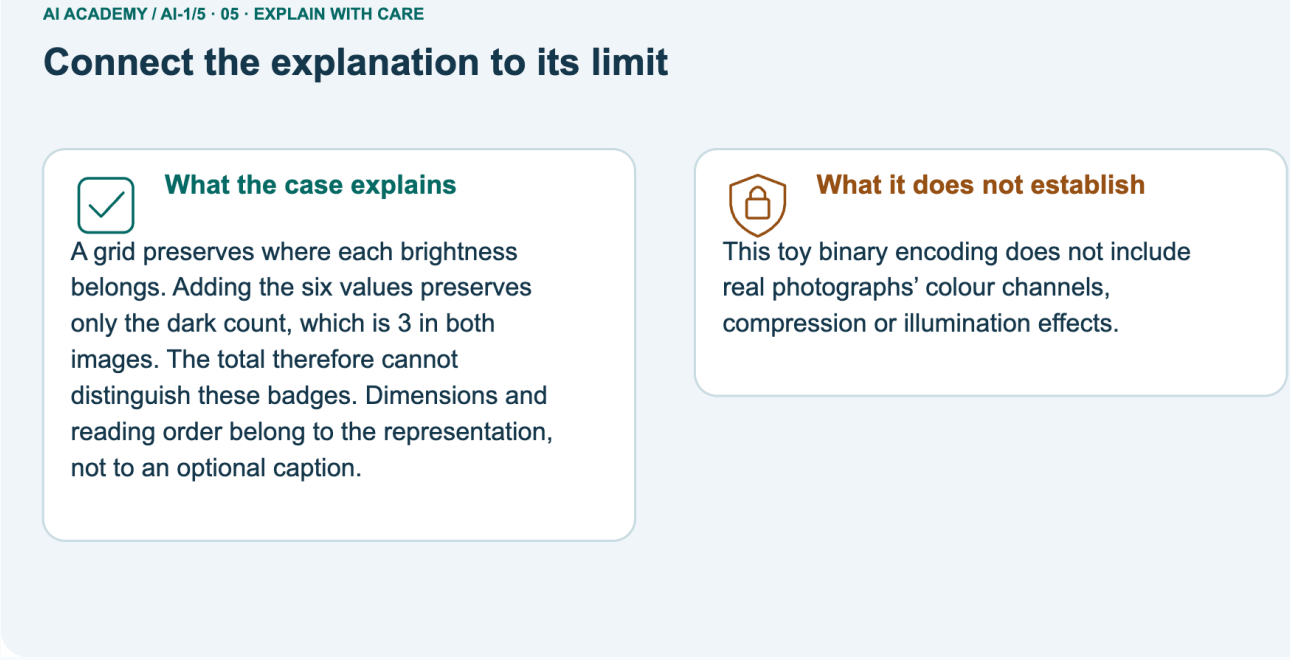
Original case: The totals agree, but middle-top and middle-bottom positions differ. Changed condition: Change A's bottom-left **pixel** from 0 to 1. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

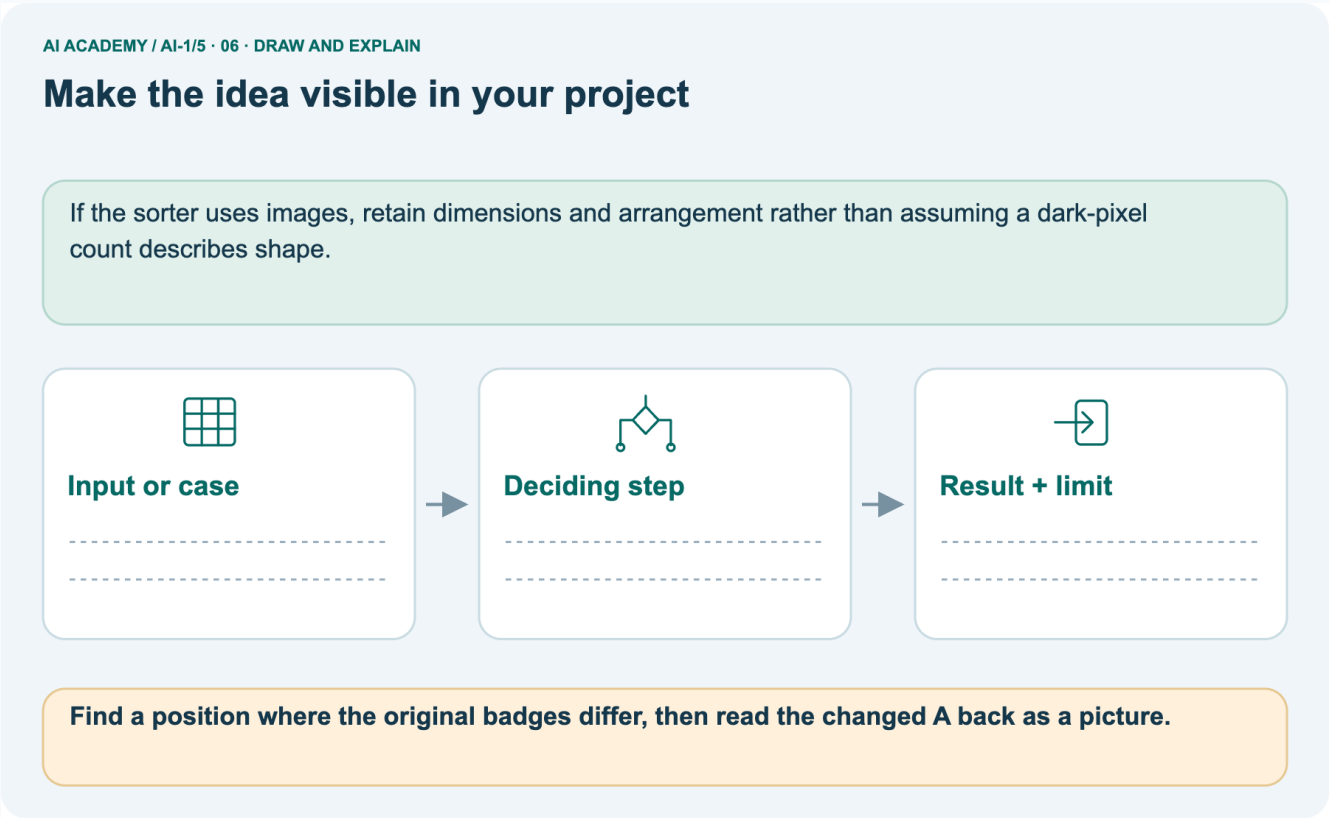
A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption. Limit: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape. Draw your **input**, deciding step, result and limit.

Supported question: Find a position where the original badges differ, then read the changed A back as a picture.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.